

# KL Divergence

Math Foundations — Concept 38

## 1 Definition

Let  $p$  and  $q$  be probability distributions on a common measurable space  $(\mathcal{X}, \mathcal{F})$ , with  $p$  absolutely continuous with respect to  $q$  (written  $p \ll q$ ). The *Kullback–Leibler divergence* of  $p$  from  $q$  is

$$D_{\text{KL}}(p \parallel q) = \mathbb{E}_p \left[ \log \frac{dp}{dq} \right].$$

**Definition 1** (Discrete KL divergence). *For probability mass functions  $p, q$  on a countable set  $\mathcal{X}$ ,*

$$D_{\text{KL}}(p \parallel q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)},$$

*with conventions  $0 \log 0 = 0$  and  $0 \log(0/0) = 0$ . The sum is  $+\infty$  if there exists  $x$  with  $p(x) > 0$  and  $q(x) = 0$ .*

**Definition 2** (Continuous KL divergence). *For probability densities  $f_p, f_q$  on  $\mathbb{R}^d$  (with respect to Lebesgue measure),*

$$D_{\text{KL}}(p \parallel q) = \int_{\mathbb{R}^d} f_p(x) \log \frac{f_p(x)}{f_q(x)} dx,$$

*defined whenever the integrand is integrable and  $\{f_q = 0\} \subseteq \{f_p = 0\}$  up to null sets.*

**Remark 3.**  $D_{\text{KL}}$  is not symmetric and does not satisfy the triangle inequality, so it is not a metric. Nonetheless it is non-negative, vanishes only when  $p = q$ , and behaves additively under independence, justifying its role as a “relative information” measure.

## 2 Gibbs’ inequality

**Theorem 4** (Non-negativity of KL). *For any pair of probability distributions  $p, q$  on the same space with  $p \ll q$ ,*

$$D_{\text{KL}}(p \parallel q) \geq 0,$$

*with equality if and only if  $p = q$  almost everywhere.*

*Proof.* Write  $r(x) = q(x)/p(x)$  on the support of  $p$ . By definition,

$$D_{\text{KL}}(p \parallel q) = \mathbb{E}_p \left[ \log \frac{p(X)}{q(X)} \right] = \mathbb{E}_p [-\log r(X)].$$

The function  $\varphi(t) = -\log t$  is strictly convex on  $(0, \infty)$ . By Jensen’s inequality applied to  $\varphi$  and the random variable  $r(X)$  under  $p$ ,

$$\mathbb{E}_p [-\log r(X)] \geq -\log \mathbb{E}_p [r(X)].$$

We compute the right-hand side:

$$\mathbb{E}_p[r(X)] = \int r(x) p(x) dx = \int \frac{q(x)}{p(x)} p(x) dx = \int q(x) dx = 1,$$

since  $q$  is a probability distribution. Therefore

$$D_{\text{KL}}(p \parallel q) \geq -\log 1 = 0.$$

Because  $\varphi$  is *strictly* convex, Jensen's inequality is an equality iff  $r(X)$  is  $p$ -almost-surely constant. Combined with  $\mathbb{E}_p[r(X)] = 1$ , this forces  $r \equiv 1$  on the support of  $p$ , i.e.  $p = q$  a.e. The discrete case is identical with integrals replaced by sums.  $\square$

**Remark 5.** *Run with  $\varphi = -\log$  replaced by an arbitrary convex  $\varphi$  with  $\varphi(1) = 0$ , the same proof yields the family of  $f$ -divergences (Csiszár). KL corresponds to  $\varphi(t) = t \log t$  (after a sign convention); other choices recover total variation, Hellinger, and  $\chi^2$  divergences.*

### 3 Consequences

**Lemma 6** (Cross-entropy decomposition). *For all  $p, q$  with  $p \ll q$ ,*

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q),$$

where  $H(p) = -\mathbb{E}_p[\log p]$  and  $H(p, q) = -\mathbb{E}_p[\log q]$ .

By Theorem ??,  $H(p, q) \geq H(p)$  with equality iff  $q = p$ . Hence minimizing cross-entropy in  $q$  for fixed  $p$  is the same as minimizing KL divergence — the foundational identity behind maximum likelihood estimation and the categorical cross-entropy loss.